Day 9 Task: Safe Malware Analysis

Part A - Create and hash the EICAR test file

Question	Your answer
SHA-256 of your eicar file	275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f
Did your antivirus react? What did it say/do?	Yes, Windows Defender blocked read access and threw the error: `"Operation did not complete successfully because the file contains a virus or potentially unwanted software"`
Why does AV flag a harmless file? (1 line)	It is an industry-standard test string created to safely verify that an antivirus engine is active and detecting threats without exposing the system to actual malware.

Part B - Look it up on Virus Total

Question	Your answer
How many engines detect it? (e.g. 60/70)	63/74
What name do engines give it? (e.g. EICAR-Test-File)	Virus/EICAR_Test_File
What file type does Virus Total report?	Text file
Under 'Details', does the SHA-256 match yours exactly?	**Yes**, it matches `131f95c51cc819465fa1797f6ccacf9d494aaaff4

Part C - Investigate an unknown hash

Question	Your answer
Is the hash known to VirusTotal? (found / not found)	Not found.
If found: how many detections and what family? If not found: what does 'not found' tell you?	**"Not found"** tells you that VirusTotal has never indexed or scanned a file with this specific SHA-256 hash before. It does **not** mean the file is safe it could be zero-day malware.
Would you upload the actual file to find out? Why or why not?	No, I would NOT upload the actual file. Uploading it would share it with researchers and potentially the attackers themselves worldwide. If the file contains sensitive company data or is a targeted attack, uploading it breaches

	confidentiality. We must analyze it in a secure, isolated, internal sandbox instead.

Part D - Read a sandbox report

Question	Your answer
What file/sample did you look at? (name or hash)	**URL Sample:** https://email-recovery.help/pages/0ed110d8-a8cd-4243-8a3b-824912a47d00/...
Name one file it created or dropped	ExpandedDomainsFilterGlobal.json (downloaded by msedge.exe from www.bing.com
Name one domain or IP it contacted (its C2?)	email-recovery.help
One persistence or registry change you noticed	Microsoft Edge (msedge.exe) created process execution entries and read/query system timezone info via WMI (WmiPrvSE.exe)

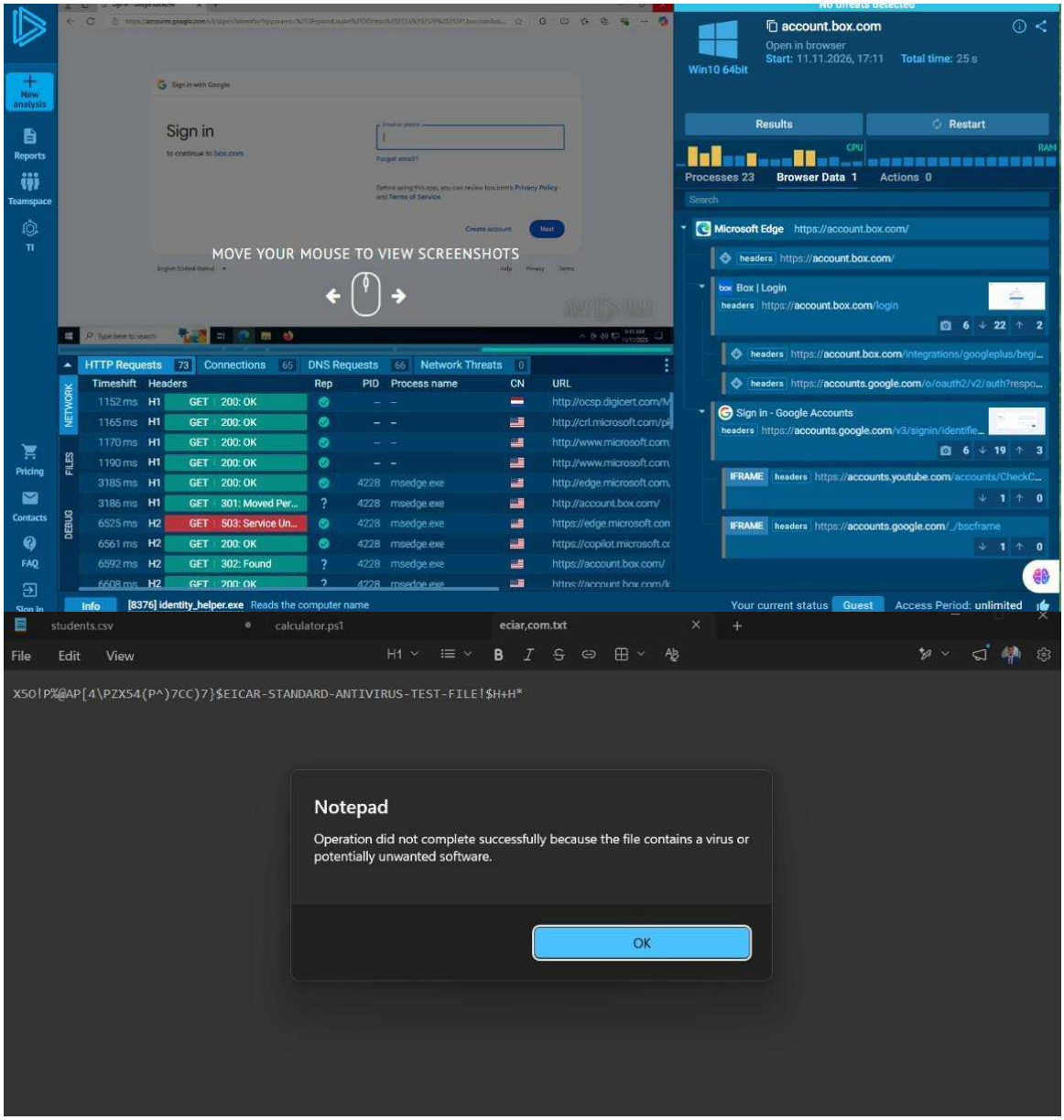
Part E - Map an attack to the kill chain

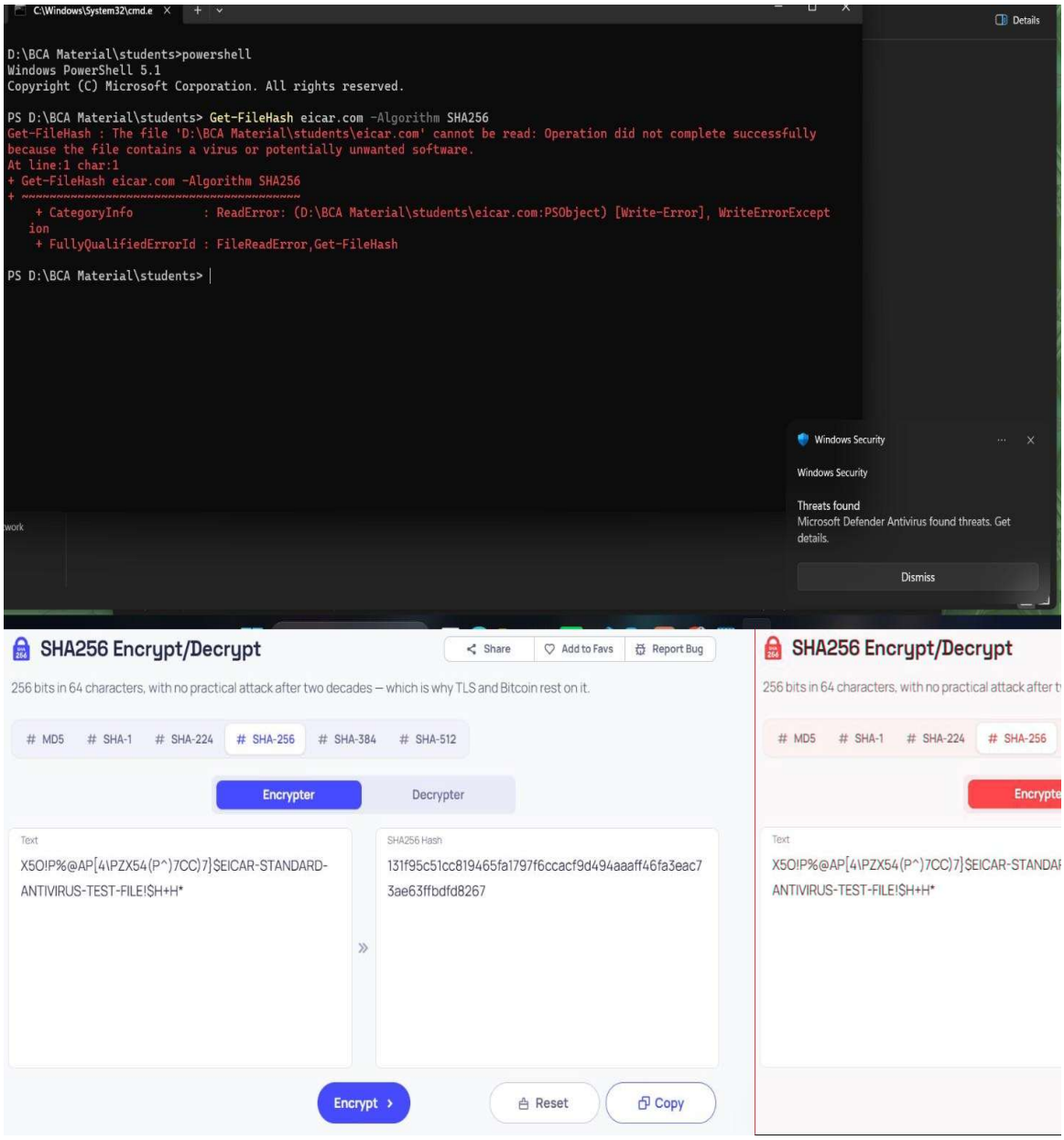
Attack chosen: WannaCry Ransomware (2017)

Stage	What happened in this attack
Recon	Attackers continuously scanned the global internet for machines with port 445 exposed and vulnerable to the EternalBlue SMB exploit.
Deliver	The malware was delivered directly over the network via the SMB port without requiring any user interaction or phishing email.
Exploit	WannaCry used the leaked NSA "EternalBlue" exploit to overflow the buffer in the Windows SMBv1 protocol, gaining system-level code execution.
Install / Persistence	The malware installed itself as a service (mssecsvc2.0) to ensure it ran silently in the background and persisted through reboots.
C2 / Control	Before encrypting, it attempted to reach an unregistered "kill switch" domain (e.g., iuqerfsodp9ifjaposdfjhgosurijfaewrwergwea.com). If it could not connect, it proceeded.
Act (impact)	It encrypted the victim's files (documents, pictures, databases) and displayed a ransom note

	demanding $300 in Bitcoin for the decryption key.

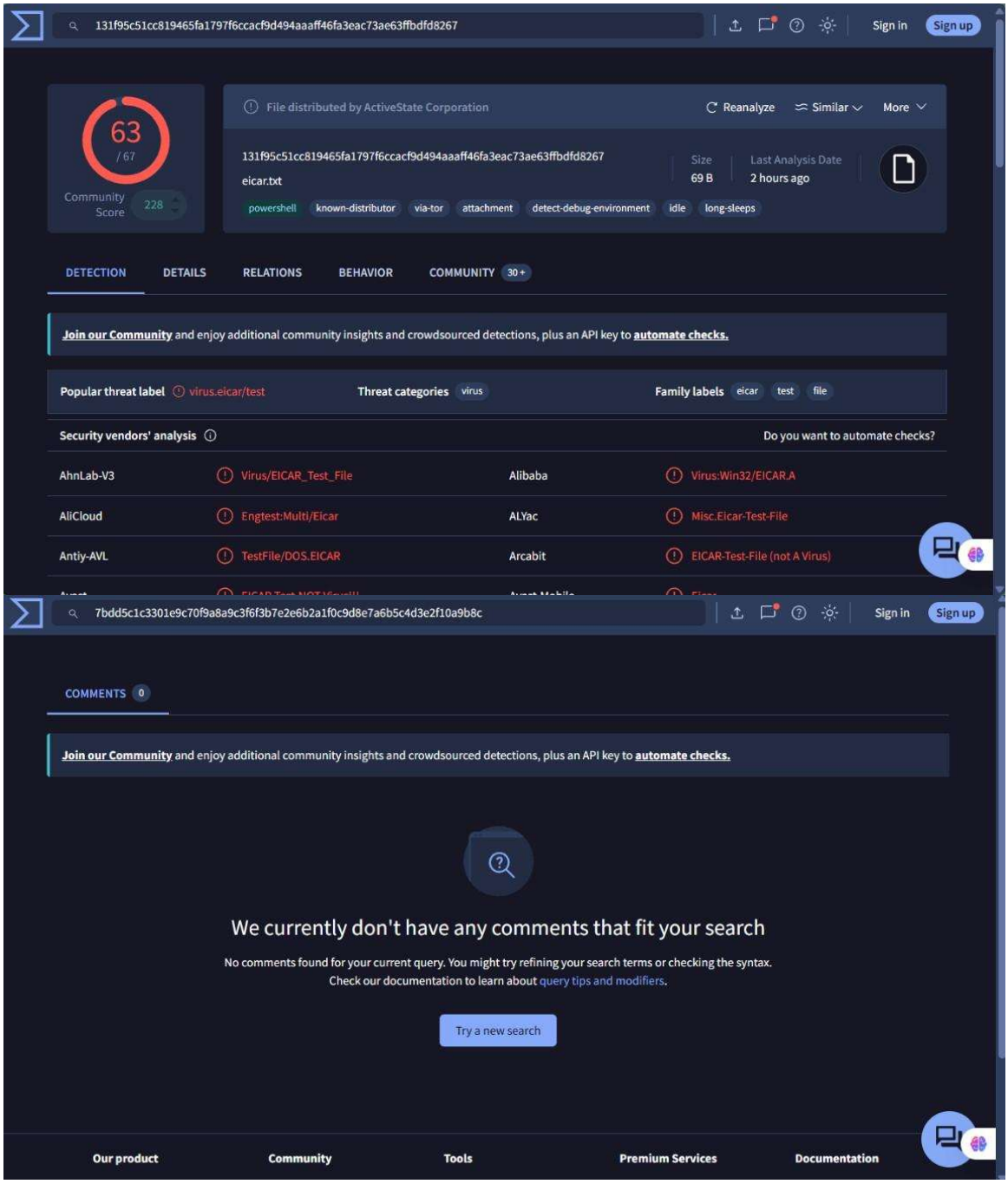
Screenshot of results:-

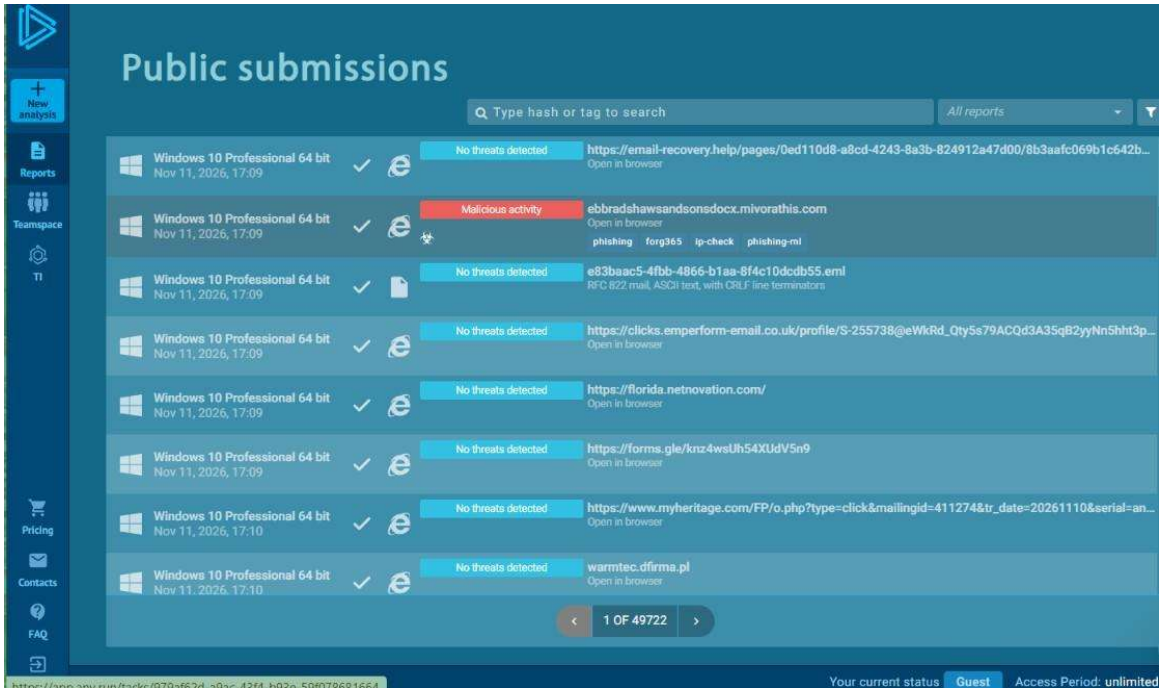



C:\Windows\System32\cmd.e
Details
D:\BCA Material\students>powershell
Windows PowerShell 5.1
Copyright (C) Microsoft Corporation. All rights reserved.
PS D:\BCA Material\students> Get-FileHash eicar.com -Algorithm SHA256
Get-FileHash : The file 'D:\BCA Material\students\eicar.com' cannot be read: Operation did not complete successfully because the file contains a virus or potentially unwanted software.
At line:1 char:1
+ Get-FileHash eicar.com -Algorithm SHA256
+ ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
+ CategoryInfo : ReadError: (D:\BCA Material\students\eicar.com:PSObject) [Write-Error], WriteErrorException
+ FullyQualifiedErrorId : FileReadError,Get-FileHash
PS D:\BCA Material\students>
Windows Security
Windows Security
Threats found
Microsoft Defender Antivirus found threats. Get details.
Dismiss

SHA256 Encrypt/Decrypt
Share
Add to Favs
Report Bug
256 bits in 64 characters, with no practical attack after two decades — which is why TLS and Bitcoin rest on it.
MD5
SHA-1
SHA-224
SHA-256
SHA-384
SHA-512
Encrypter
Decrypter
Text
X5O!P%@AP[4\PZX54(P^)7CC)7]$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*
SHA256 Hash
131f95c51cc819465fa1797f6ccacf9d494aaaff46fa3eac73ae63ffbdfd8267
Encrypt
Reset
Copy

SHA256 Encrypt/Decrypt
MD5
SHA-1
SHA-224
SHA-256
Text
X5O!P%@AP[4\PZX54(P^)7CC)7]$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*


131f95c51cc819465fa1797f6ccacf9d494aaaff46fa3eac73ae63ffbdfd8267
Sign in
Sign up
63
/ 67
Community Score
228
File distributed by ActiveState Corporation
Reanalyze
Similar
More
131f95c51cc819465fa1797f6ccacf9d494aaaff46fa3eac73ae63ffbdfd8267
eicar.txt
Size
69 B
Last Analysis Date
2 hours ago
powershell
known-distributor
via-tor
attachment
detect-debug-environment
idle
long-sleeps
DETECTION
DETAILS
RELATIONS
BEHAVIOR
COMMUNITY 30+
Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.
Popular threat label virus.eicar/test
Threat categories virus
Family labels eicar test file
Security vendors' analysis
Do you want to automate checks?
AhnLab-V3
Virus/EICAR_Test_File
Alibaba
Virus:Win32/EICAR.A
AliCloud
Engtest:Multi/Eicar
ALYac
Misc.Eicar-Test-File
Antiy-AVL
TestFile/DOS.EICAR
Arcabit
EICAR-Test-File (not A Virus)
7bdd5c1c3301e9c70f9a8a9c3f6f3b7e2e6b2a1f0c9d8e7a6b5c4d3e2f10a9b8c
Sign in
Sign up
COMMENTS 0
Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.
We currently don't have any comments that fit your search
No comments found for your current query. You might try refining your search terms or checking the syntax.
Check our documentation to learn about query tips and modifiers.
Try a new search
Our product
Community
Tools
Premium Services
Documentation


Public submissions
New analysis
Reports
Teamspace
TI
Pricing
Contacts
FAQ
Type hash or tag to search
All reports
Windows 10 Professional 64 bit
Nov 11, 2026, 17:09
No threats detected
https://email-recovery.help/pages/0ed110d8-a8cd-4243-8a3b-824912a47d00/8b3aafc069b1c642b...
Open in browser
Windows 10 Professional 64 bit
Nov 11, 2026, 17:09
Malicious activity
ebbradshawsandsonsdocx.mivorathis.com
Open in browser
phishing
forg365
ip-check
phishing-ml
Windows 10 Professional 64 bit
Nov 11, 2026, 17:09
No threats detected
e83baac5-4fbb-4866-b1aa-8f4c10dcdb55.eml
RFC 822 mail, ASCII text, with CRLF line terminators
Windows 10 Professional 64 bit
Nov 11, 2026, 17:09
No threats detected
https://clicks.emperform-email.co.uk/profile/S-255738@eWkRd_Qty5s79ACQd3A35qB2yyNn5hht3p...
Open in browser
Windows 10 Professional 64 bit
Nov 11, 2026, 17:09
No threats detected
https://florida.netnovation.com/
Open in browser
Windows 10 Professional 64 bit
Nov 11, 2026, 17:09
No threats detected
https://forms.gle/knz4wsUh54XUdV5n9
Open in browser
Windows 10 Professional 64 bit
Nov 11, 2026, 17:10
No threats detected
https://www.myheritage.com/FP/o.php?type=click&mailingid=411274&tr_date=20261110&serial=an...
Open in browser
Windows 10 Professional 64 bit
Nov 11, 2026, 17:10
No threats detected
warmtec.dfirma.pl
Open in browser
1 OF 49722
Your current status
Guest
Access Period: unlimited